MSC DATA SCIENCE · INDIVIDUAL PROJECT

Predicting Automobile Fuel Efficiency Using Machine Learning

Siddhartha Ragipani ·

Student ID: 24074708

Dataset: [UCI Auto MPG Dataset]

Github: https://github.com/ragipanisiddhartha/auto-mpg-fuel-efficiency-ml/tree/main

Date: March 28, 2026

University of Hertfordshire

Supervisor: Sunina Sharvy

ROADMAP

Presentation Roadmap

- Project motivation & research question
- Data & preprocessing
- EDA & key relationships
- Models, results & residuals
- Discussion, limitations & future work

BACKGROUND

Problem Statement & Research Question

Context

- Fuel efficiency is a critical factor for environmental sustainability and cost reduction
- Accurate mpg prediction helps manufacturers optimize vehicle design
- Consumers benefit from informed purchasing decisions

Problem Statement

Given basic technical attributes of a vehicle such as (engine specifications, weight, year, origin), can we **accurately predict its fuel efficiency**?

Research Question

"How accurately can machine learning models predict a car's mpg from its technical attributes, and which regression model performs best on the Auto MPG dataset?"

SCOPE & DATA

Project Objectives & Dataset Overview

Main Objectives

Explore the Auto MPG dataset through comprehensive Exploratory Data Analysis (EDA)

Build baseline and advanced regression models to predict mpg

Compare linear, regularized, and tree-based models using RMSE and R^2

Optimize models through hyperparameter tuning and cross-validation

Interpret the best-performing model and relate findings to published literature

Auto MPG Dataset — UCI ML Repository

398 Vehicles · 1970–1982

Target: **mpg** (miles per gallon) · Publicly available benchmark

Features: cylinders, displacement, horsepower, acceleration, weight, model_year, origin

	mpg	cylinders	displacement	weight	acceleration	model_year	origin
count	398.000000	398.000000	398.000000	398.000000	398.000000	398.000000	398.000000
mean	23.514573	5.454774	193.425879	2970.424623	15.568090	76.010050	1.572864
std	7.815984	1.701004	104.269838	846.841774	2.757689	3.697627	0.802055
min	9.000000	3.000000	68.000000	1613.000000	8.000000	70.000000	1.000000
25%	17.500000	4.000000	104.250000	2223.750000	13.825000	73.000000	1.000000
50%	23.000000	4.000000	148.500000	2803.500000	15.500000	76.000000	1.000000
75%	29.000000	8.000000	262.000000	3608.000000	17.175000	79.000000	2.000000
max	46.600000	8.000000	455.000000	5140.000000	24.800000	82.000000	3.000000

The dataset includes 398 vehicles with complete data across all variables. Fuel efficiency varies widely, with mpg ranging from 9.0 to 46.6 and averaging 23.5, while weight is the heaviest feature at a mean of 2,970 lbs and a range of 1,613–5,140 lbs. Cylinders span 3 to 8 with a mean of 5.5, model year ranges from 1970 to 1982 (coded 70–82), and origin is coded 1–3 to represent geographic regions. Acceleration averages 15.6 seconds with relatively low variability, providing a clear statistical foundation for the modeling task ahead.

FOUNDATION & PREPARATION

Literature Review & Data Preprocessing

Key Literature Findings

Regression on Auto MPG

- Strong negative correlation: engine size, weight → mpg
- Linear models baseline $R^2 \approx 0.75–0.80$
- Model year shows consistent positive impact

Ensemble & Evaluation Methods

Random Forest & Gradient Boosting outperform linear models on tabular data *(Breiman, 2001)*

- RMSE gives intuitive error in mpg units; R^2 quantifies variance explained
- Cross-validation ensures generalization

James et al. (2021) *An Introduction to Statistical Learning*, Springer · Dua & Graff (2019) UCI ML Repository · Breiman (2001) *Machine Learning*, 45(1), 5–32.

Data Preprocessing Steps

Missing Data Handling

6 rows with missing horsepower (marked '?') → converted to NAN → removed · Final **rows= 392**

Data Type Validation

float64 for continuous variables, int64 for categorical · Verified no remaining null values

Feature Engineering

Dropped `car_name` (identifier, not predictive) · Retained all 7 predictive features · No initial scaling (tree models don't require it)

Prior work on the Auto MPG dataset and *An Introduction to Statistical Learning* show that linear models typically reach $R^2 \approx 0.75–0.80$, while tree-based ensembles such as Random Forest and Gradient Boosting often perform better on tabular regression tasks, which motivates our choice of models. (James et al., 2021; Breiman, 2001; Dua & Graff, 2019)
These studies also consistently report strong negative effects of vehicle weight and engine size (displacement, cylinders) on mpg, and our EDA reproduces the same patterns on the UCI Auto MPG dataset. (Dua & Graff, 2019; James et al., 2021)

EXPLORATORY DATA ANALYSIS

EDA: Feature Distributions & Key Relationships

Distribution Highlights

MPG

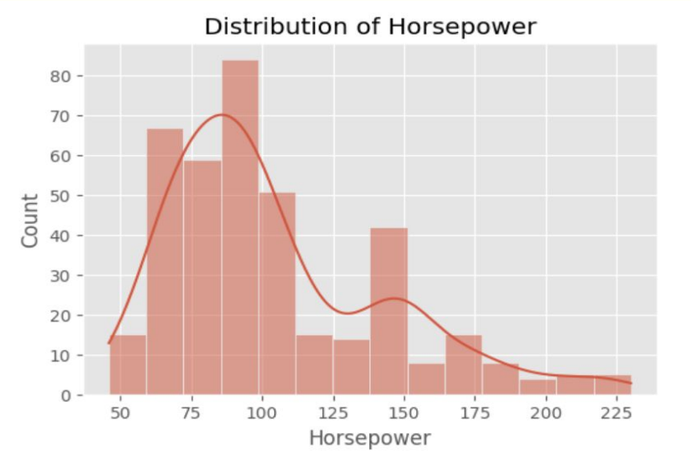
Range 9–46.6 mpg
Median ~23
Right-skewed

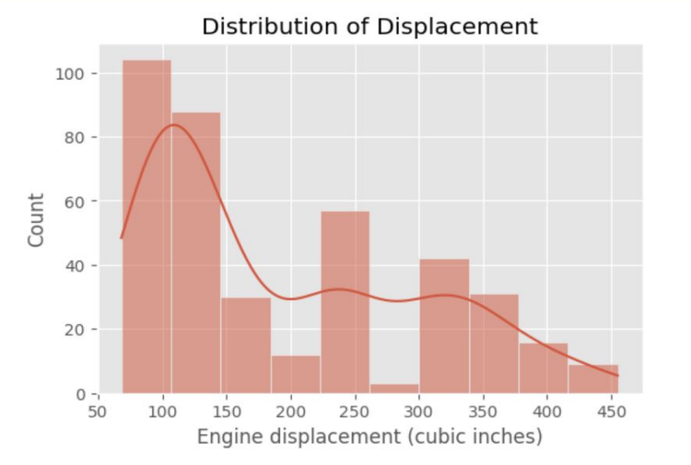
Horsepower & Displacement

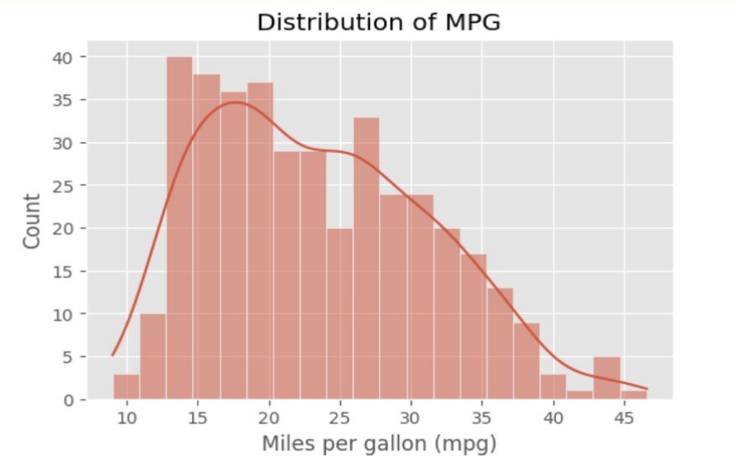
Right-skewed
Most vehicles 80–120 hp
Few outliers

Model Year

Even distribution 1970–1982
13 years of evolution







These three histograms summarise how fuel efficiency and engine size are distributed in the dataset. The mpg plot shows a right-skewed distribution, with most cars achieving low-to-mid 20s mpg and a smaller tail of highly efficient vehicles. The displacement and horsepower plots are also right-skewed, indicating many vehicles with moderate engines and relatively few high-displacement, high-horsepower outliers.

EXPLORATORY DATA ANALYSIS - CONTINUED

EDA: Key Correlations (Continued)

Key Correlations with MPG

Strong Negative ↓

Weight: r ≈ −0.83 · Heavier = less efficient

Displacement: r ≈ −0.81 · Larger engines worse

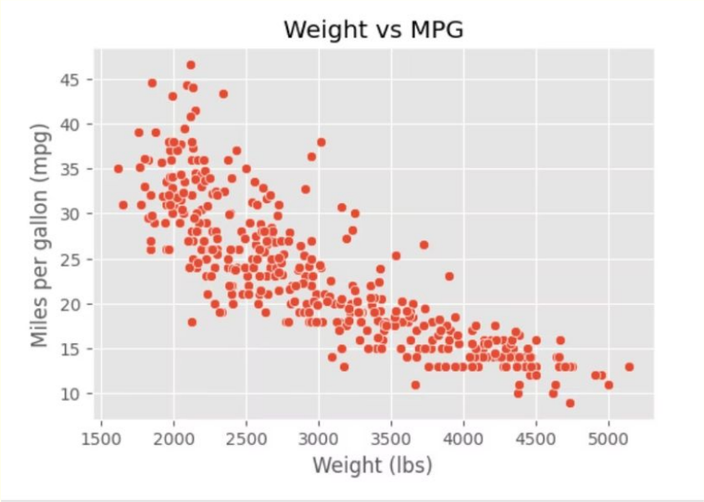
Cylinders / Horsepower: r ≈ −0.78

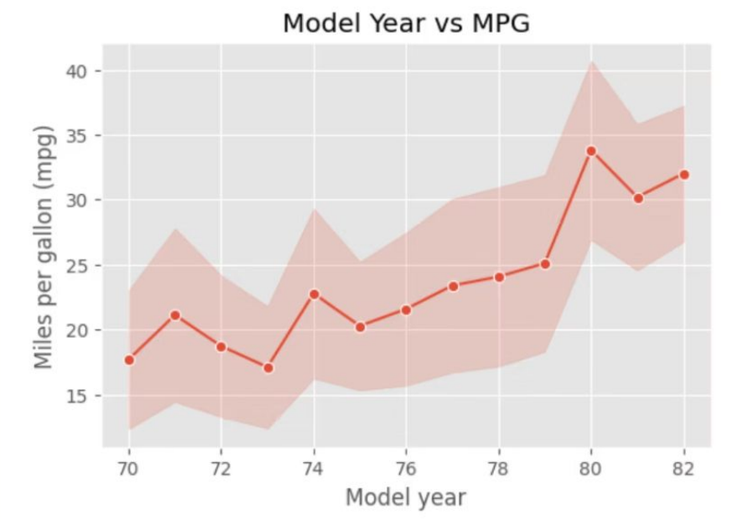
Positive ↑

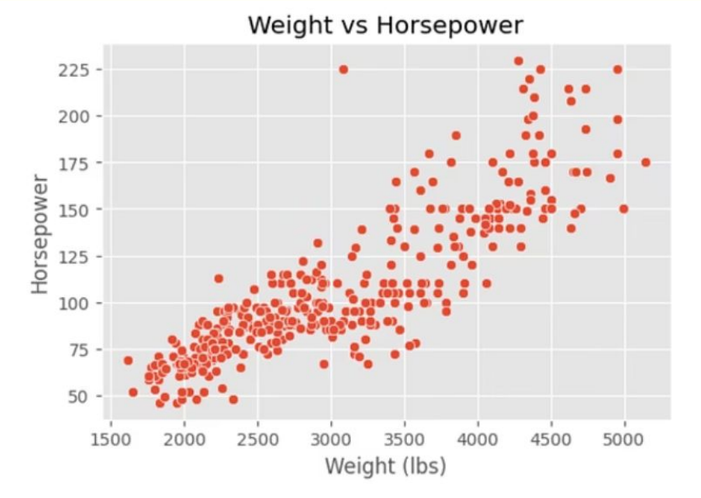
Model Year: r ≈ +0.58 · Newer cars more efficient

Origin: r ≈ +0.57 · EU/JP more efficient than US

Acceleration: r ≈ +0.42 · Slower accel → better mpg







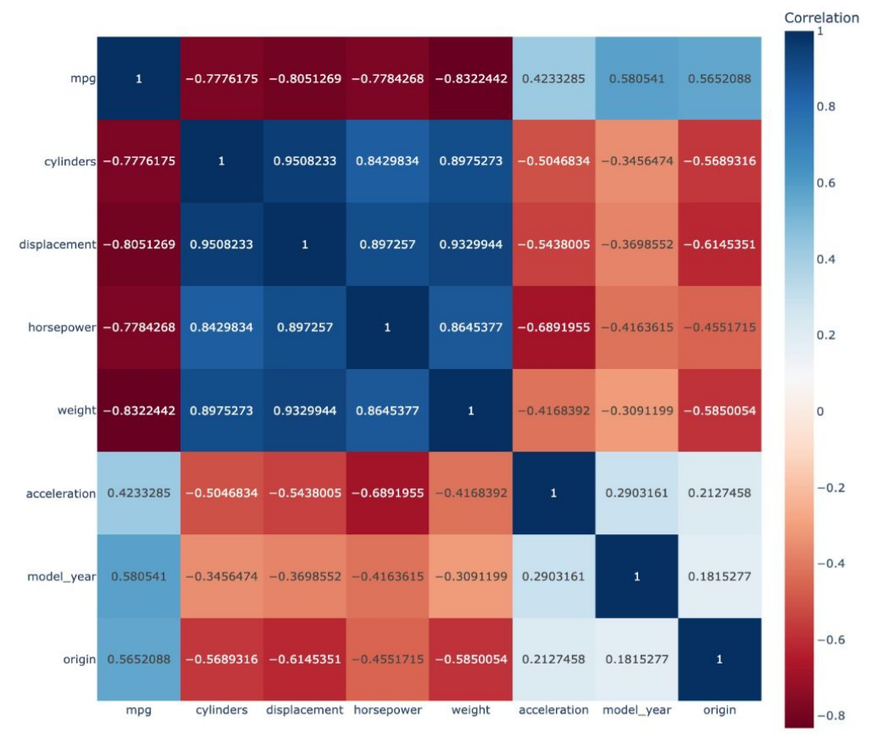
These three plots illustrate the key relationships behind your correlation summary. The weight–horsepower scatterplot shows that heavier cars almost always come with higher horsepower, confirming strong positive dependence between engine size and power. The model-year line plot shows average mpg increasing steadily from the early 1970s to the early 1980s, so newer cars are clearly more fuel-efficient on average. Finally, the weight–mpg scatterplot shows a strong downward trend: as vehicle weight increases, mpg drops sharply, which matches the large negative correlation between weight and fuel efficiency.

METHODOLOGY

Correlation Analysis & Modeling Strategy

Multicollinearity Among Features

Engine-related features are strongly correlated with each other, with cylinders, displacement, and horsepower all clustering around 0.90–0.95. MPG is negatively correlated with weight (-0.83), displacement (-0.81), cylinders (-0.78), and horsepower (-0.78), while model_year (+0.58) and origin (+0.56) show positive relationships;



EXPERIMENTAL SETUP

Data Split & Evaluation:

80% train / 20% test, Fixed seed (42), 5-fold cross-validation, Metrics: **RMSE** (mpg) and **R^2**

Models Tested:

- Linear Baseline: Multiple Linear Regression
- Regularized: Ridge Regression, Lasso Regression
- Tree-Based: Decision Tree, Random Forest

Hyperparameter Tuning:

- Ridge/Lasso: GridSearchCV for alpha
- Decision Tree: max_depth, min_samples_leaf, min_samples_split
- Random Forest: n_estimators, max_depth, max_features

Correlation insights :

Weight ↔ Displacement: r ≈ +0.93 & Cylinders ↔ Displacement: r ≈ +0.95
Horsepower ↔ Displacement: r ≈ +0.90

RESULTS

Model Performance Comparison

Cross-Validation Results — All Models

Model	CV RMSE	CV R^2
Linear Regression	3.49 mpg	0.80
Ridge Regression	3.49 mpg	0.80
Lasso Regression	3.49 mpg	0.80
Decision Tree	3.84 mpg	0.75
Random Forest ✓	**2.97 mpg**	**0.85**

Key Findings

Linear Baseline

Test RMSE: **3.10 mpg** · R^2: **0.84** · Good fit, but residuals suggest some non-linearity

Ridge / Lasso

Matches baseline performance · Mainly improves coefficient stability

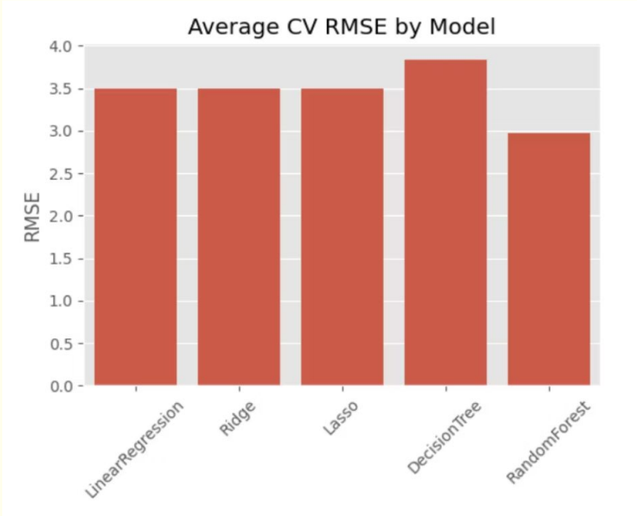
Random Forest — Best

Lowest RMSE (**2.97 mpg**) · Highest R^2 (**0.85**) · ~15% better than the linear baseline

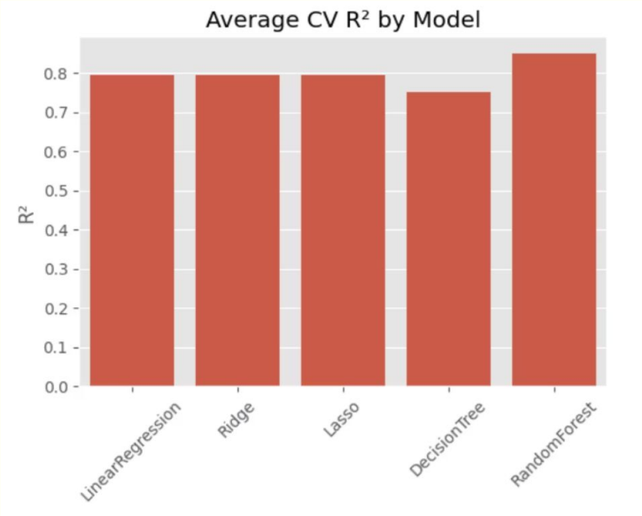
RESULTS - CONTINUED

Model Performance Comparison (Continued)

COMPARISON CHARTS



CV RMSE comparison



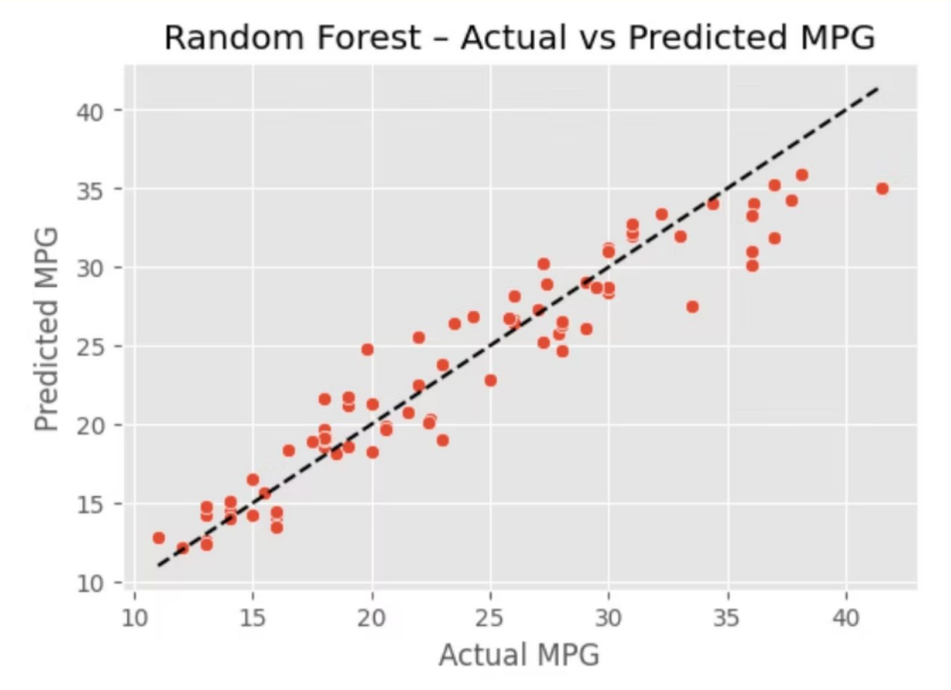
CV R^2 comparison

Linear Regression, Ridge, and Lasso all deliver nearly identical CV RMSE (~3.49 mpg) and CV R^2 (0.80), indicating that regularization has minimal impact here. Decision Tree performs worse with higher RMSE (3.84 mpg) and lower R^2 (0.75). Random Forest clearly leads with the lowest RMSE (2.97 mpg) and highest R^2 (0.85), showing about a 15% RMSE reduction and 5% R^2 gain versus the linear baseline and confirming the strength of ensemble methods for this task.

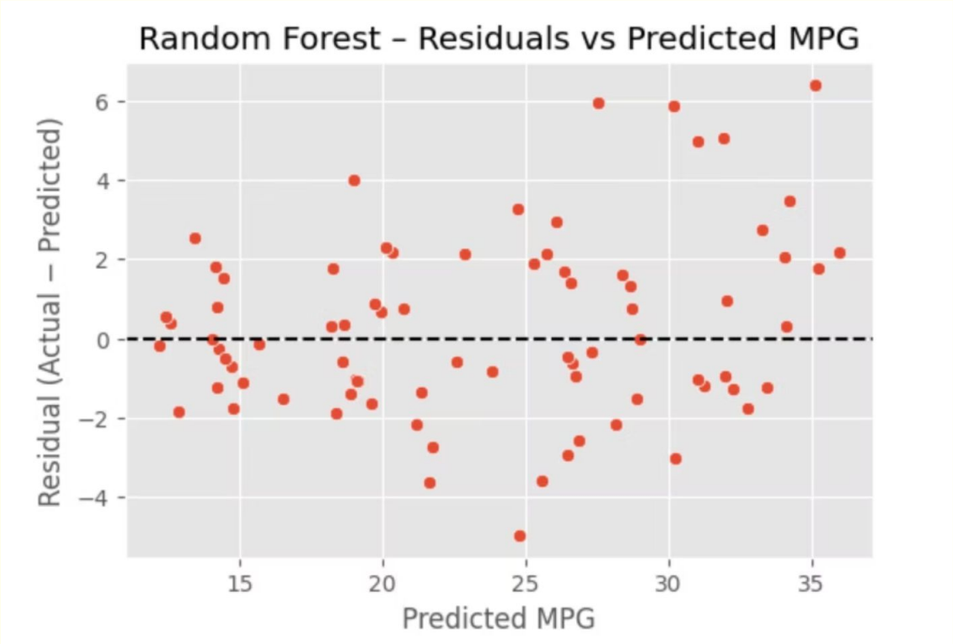
RESULTS - CONTINUED

Model Diagnostics (Continued)

MODEL DIAGNOSTICS



Random Forest – Actual vs Predicted MPG



Random Forest – Residuals vs Predicted MPG - Residuals are mostly centred around zero with no strong pattern, indicating a good fit with some remaining variability at higher MPG.

BEST MODEL

Optimized Random Forest Performance

Tuned Hyperparameters

n_estimators: **300**
max_depth: **10**
min_samples_leaf: **1**
max_features: **0.5 (sqrt)**
random_state: **42**

Final Test Set Performance

2.28
RMSE (mpg)

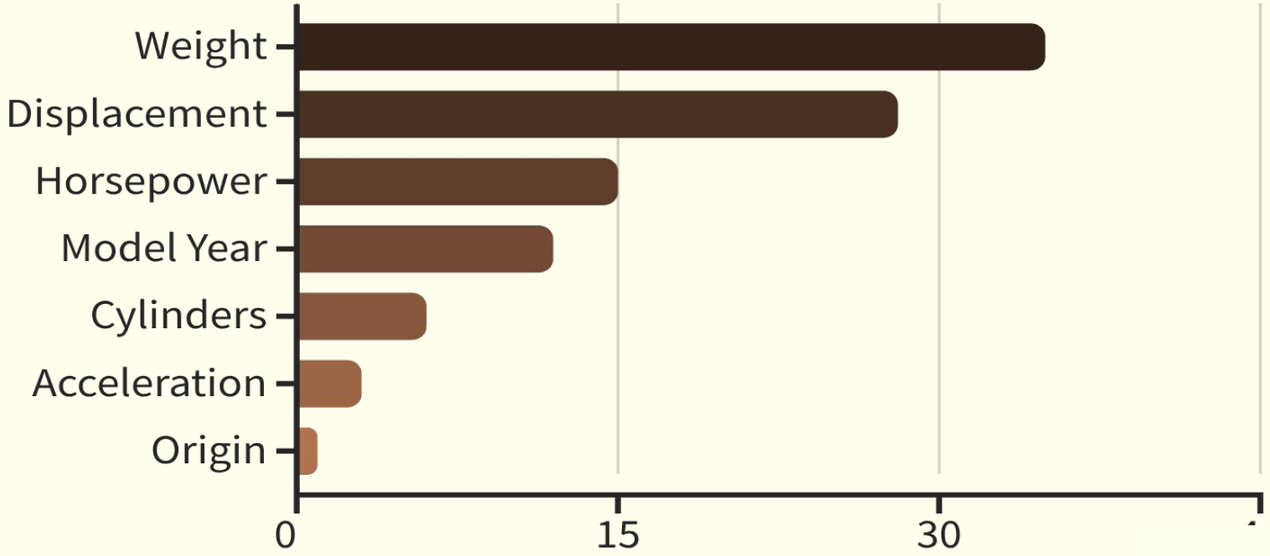
0.91
R^2 Score

Improvement Over Linear Baseline

26%
RMSE Reduction
3.10 → 2.28 mpg

8%
R^2 Increase
0.84 → 0.91

Feature Importance (Top 7)



DISCUSSION & LIMITATIONS

Analysis, Limitations & Future Work

Why Random Forest Outperforms Linear Models

Non-Linear Relationships

Captures complex interactions between weight, engine size, and efficiency

Robust to Multicollinearity

Handles correlated features (e.g., weight ↔ displacement: +0.93) effectively

Outlier Resistance

Ensemble averaging reduces impact of anomalous data points

Current Limitations

Data from **1970–1982** (44 years old) · Only 392 observations · May not generalize to modern hybrids, EVs, or turbocharged engines

🔧 Missing features: aerodynamics, transmission type, tire specs · No real-world driving conditions · Single dataset evaluation only

Future Directions

Modern Datasets: Incorporate 2020+ vehicle data with hybrid/EV technology

Advanced Models: Test XGBoost, LightGBM, Neural Networks

Feature Engineering: Add interaction terms, polynomial features

Interpretability: Apply SHAP values for detailed feature impact analysis

Emissions Modelling: Extend to predict CO_2 alongside mpg

CONCLUSION

Conclusion & Key Takeaways

1

✓ Research Question Answered

Random Forest achieves **RMSE = 2.28 mpg** and **R^2 = 0.91**, outperforming linear baseline (RMSE 3.10, R^2 0.84) by **26%**

2

Systematic Comparison

5 models evaluated with proper cross-validation and hyperparameter tuning · Decision Trees underperform; regularized models match baseline

3

⚖️ Ethics & Reproducibility

Public UCI benchmark · No personal data · Full code on GitHub · Transparent evaluation · Positive societal impact for vehicle design & emissions policy

Main Findings

Weight (35%) and **displacement** (28%) are dominant predictors — aligning with domain knowledge and literature

- Tree-based ensemble methods excel for tabular regression tasks
- Methodology is applicable to modern vehicle datasets and sustainability challenges

---

Thank you — Questions? · Contact: ragipanisiddhartha008@gmail.com